

Predicting Singer Gender from Song Lyrics: A Controlled Comparison Between Sparse Lexical and Contextual Transformer Models

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Abstract

This paper investigates the task of predicting singer gender from English song lyrics using a controlled supervised text classification framework. We compare two distinct modeling paradigms: a sparse lexical baseline based on TF-IDF representations paired with a linear classifier, and a contextual neural approach based on a fine-tuned BERT model. To prevent identity leakage, we enforce a strict artist-level split, ensuring that songs by the same artist never appear across multiple data partitions. Our experiments on a curated balanced corpus show that the contextual model significantly outperforms the lexical baseline across all major evaluation metrics. Beyond aggregate performance, we also analyze the discriminative lexical features learned by the linear model and examine systematic error patterns that reveal the limits of text-only gender inference.

1 Introduction

Song lyrics constitute a rich domain for natural language processing because they combine recurring lexical patterns, stylistic conventions, and identity-related signals. Prior work has shown that lyrical content can be used to study thematic structure, bias and social representation. In particular, computational studies have increasingly focused on measuring gender bias and sexism within song lyrics through topic modeling and quantitative analysis (Chen et al., 2024b,a; Arango et al., 2022). Beyond thematic content, some research in the digital humanities has demonstrated that lyrical structures and rhyme patterns can efficiently be leveraged for author profiling and attribution tasks (Yilmaz and Scheffler, 2023). In parallel, recent NLP efforts have advanced toward detecting and mitigating these implicit gender biases directly within contextual word embeddings (Zhang et al., 2024).

In this work, we ask whether a singer’s gender can be predicted from lyrics and whether a con-

textual transformer model provides a measurable advantage over a transparent lexical baseline.

We make three main contributions:

- We construct a curated lyric dataset with binary gender labels aligned through external metadata sources.
- We compare a classical TF-IDF + linear classifier pipeline with a fine-tuned BERT model under identical evaluation conditions.
- We provide an interpretability and error analysis of the lexical baseline and the transformer model to better understand what linguistic cues are actually being captured.

2 Dataset and Task Definition

Our corpus is derived from a large public [dataset](#) enriched with metadata from Spotify ($N = 955,320$). The initial collection contains song-level information and raw lyrics, but does not directly provide singer gender labels. We therefore perform an enrichment step to align artist names with demographic metadata. To achieve this, we programmatically queried the MusicBrainz API with the `musicbrainzngs` Python library, an open-source, community-maintained encyclopedic music database. For each unique artist name in our dataset, a search query was executed to retrieve their corresponding metadata profile, and more specifically their gender. The API allowed us to filter entries by entity type, separating collective entities (which is classified as "Group") from individual artists (which is classified as "Person"). For individual performers, we extracted the specific demographic attribute gender. For computational efficiency, a random subsample of 50,000 songs was initially selected for processing.

Through this [notebook](#), we successfully mapped textual artist strings to verified demographic pro-

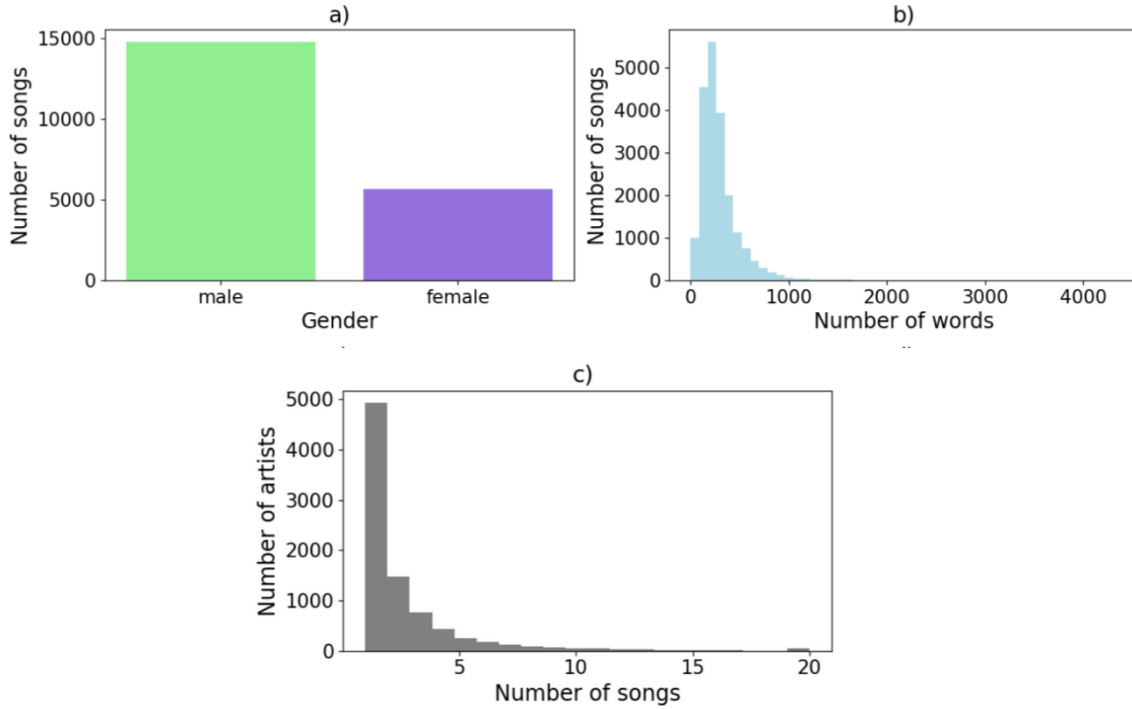


Figure 1: Dataset distribution characteristics: a) Number of songs by gender (male vs. female), b) Histogram of number of words for male artists, c) Histogram of number of words for female artists, and d) Histogram of number of songs per artist.

files, allowing us to construct a supervised binary classification task.

2.1 Filtering Criteria

We apply several filtering criteria:

- Songs with missing lyrics, missing artist information, or exact duplicates are removed.
- Non-English content is filtered out to ensure a linguistically coherent dataset.
- We exclude bands and musical groups because their gender composition is not straightforward to encode as a single label.
- Artists whose gender is neither male nor female are excluded (Figure 1.a).

After cleaning and alignment, we obtain a large and highly curated corpus of 22,500 songs with a number of words predominantly between 0 and 1000 (Figure 1.b.). The resulting dataset contains songs from 8,569 unique artists (Figure 1.c) and is suitable for controlled experimentation. The task is defined as binary classification, where the input is the lyrics of a song and the output is the singer’s gender.

We then proceeded to the core coding of our project in the second [notebook](#).

3 Experimental Setup

To ensure a rigorous evaluation, we split the data at the artist level rather than the song level. Artists are first grouped by normalized artist names and static gender labels, then partitioned into training, validation, and test sets using stratified sampling to preserve class balance (see Figure 2). All songs associated with a given artist are assigned to the same split. We also verify that no artist appears in more than one partition.

This setup ensures that performance reflects generalization to unseen artists rather than memorization of artist-specific lexical signatures. The final evaluation is conducted on a test set containing songs from artists never seen during training or validation.

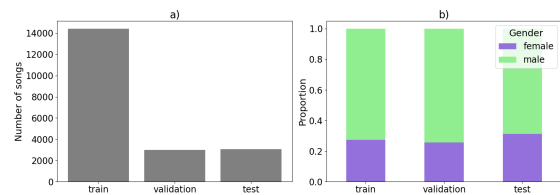


Figure 2: Data partition sizes and class balance: a) Total number of songs in train, validation, and test splits, and b) Class proportions of male and female labels within each split.

4 Preprocessing

Because the two modeling approaches rely on different assumptions about text representation, we adopt two preprocessing pipelines.

For the TF-IDF baseline, lyrics are normalized aggressively. We convert all text to lowercase, remove structural annotations such as section markers and repetition cues, strip punctuation and non-alphabetic characters, and remove standard English stopwords (Zhang et al., 2024). We intentionally keep gendered pronouns such as *he*, *she*, *him*, *her*, *his*, and *hers*, since they may carry predictive information relevant to the classification task.

For BERT, we apply a lighter preprocessing strategy. We remove only obvious structural annotations, while preserving capitalization, punctuation, and word order, since transformer models benefit from these surface cues (Zhang et al., 2024; Gonen and Goldberg, 2019). The text is then tokenized using the WordPiece tokenizer associated with the pretrained model. For computational reasons, we truncate sequences to the first 512 subword tokens, which corresponds to the standard input limit of BERT (Gonen and Goldberg, 2019).

5 Models

5.1 TF-IDF Baseline

Our baseline consists of a TF-IDF vectorizer followed by a linear classifier. The input lyrics are transformed into sparse weighted vectors using term frequency-inverse document frequency representation (Zhang et al., 2024). We evaluate two standard linear classifiers: Logistic Regression and Linear Support Vector Classification. Because the dataset is not perfectly balanced at the class level, we use class weighting to reduce majority-class bias.

Hyperparameters are selected on the validation set using grid search and optimizing the macro-F1 score. Macro-F1 score is the mean between the F1-score of men and the F1-score of women, more relevant than the accuracy in a case of imbalanced classes. The search space includes the maximum vocabulary size, n-gram range, and regularization strength.

The best-performing configuration on validation macro-F1 uses a linear classifier trained on TF-IDF vectors with both unigrams and bigrams. The optimal preprocessing configuration excluded terms appearing in fewer than 5 songs or more than 90% of the corpus, while applying sublinear scaling to

mitigate word repetition effects, which makes sense in the context of the chorus part of the song.

5.2 Fine-Tuned BERT

Our second model is a fine-tuned BERT classifier based on the bert-base-uncased checkpoint (Devlin et al., 2019). The final hidden representation of the [CLS] token is used as an aggregate sequence embedding, which is passed through a dropout layer and a linear classification head. We train the model using AdamW with weight decay and a linear learning rate schedule.

The BERT model is trained on the same artist-level split as the baseline, ensuring a fair comparison (Arango et al., 2022). Because the model is computationally more expensive than the linear baseline, we use a limited number of epochs and rely on validation-based stopping to select the best checkpoint. The validation loss grows after the second epoch, suggesting an overfitting during the training phase (Figure 4.a), but the accuracy and the macro-F1 validation are still growing after epoch 3 (Figure 4.b), showing that the last epoch is relevant to improve our model.

It should be noted that the accuracy and the macro-F1 could be improved by, for example, reducing the learning rate to reduce overfitting. However, because the present model was already better than the TF-IDF one, because the objective was to compare both of them and because of computational reasons, we decided to keep the model as it is.

The TF-IDF + Linear Classifier achieves 76.81 accuracy and 72.36 macro-F1 for the validation dataset, while Fine-Tuned BERT reaches 80.52 accuracy and 77.24 macro-F1.

6 Results

Globally, both models performed well on classifying male artists but showed lower efficacy when classifying female artists (Figure 6), suggesting that the majority class bias was not entirely eliminated.

Table 1 reports the final test-set performance of both models. The fine-tuned BERT model outperforms the TF-IDF baseline on accuracy, precision and macro-F1.

The TF-IDF + Linear Classifier achieves 64.06% accuracy and 59.69% macro-F1, while Fine-Tuned BERT reaches 68.91% accuracy and 64.07% macro-F1 (Figure 5). To assess whether this im-

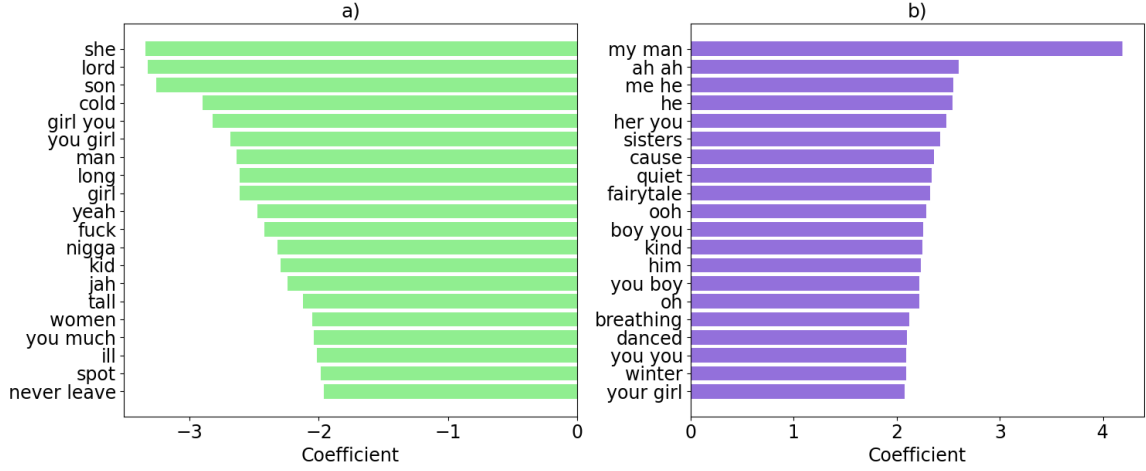


Figure 3: Most informative sparse lexical features learned by the TF-IDF linear baseline. The charts display the highest-magnitude coefficients driving the classification toward a) male singer predictions (negative coefficients) and b) female singer predictions (positive coefficients).

Model	Acc.	Prec. Macro	Rec. Macro	F1 Macro	F1 Weighted	Prec. Male	Rec. Male	Prec. Female	Rec. Female
TF-IDF + Linear	0.6406	0.5951	0.6015	0.5969	0.6464	0.7545	0.7062	0.4358	0.4969
BERT	0.6891	0.6400	0.6414	0.6407	0.6899	0.7759	0.7692	0.5041	0.5136

Table 1: Final test-set evaluation results across baseline and transformer architectures.

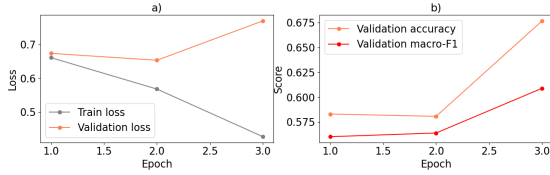


Figure 4: Fine-tuned BERT training dynamics across epochs: a) Training vs. validation loss curve showing convergence, and b) Validation accuracy and validation Macro-F1 progression.

provement is statistically meaningful, we apply McNemar’s test on paired predictions from the two models.

The improvement is especially important for the Precision Female (Prec. female). Indeed, it reveals a major performance gap, rising from 43.5 percent with TF-IDF to 50.4 percent with BERT. One can argue that the TF-IDF Baseline suffers from a high false-positive rate, because it operates purely on word frequencies, and so cannot understand syntax. When a male artist frequently uses words like "she", "her", or "girl", the model misinterprets these target tokens as structural indicators of a female author.

This potentially suggests that BERT’s superiority relies on its architecture’s ability to move beyond isolated token frequencies and capture contextual dependencies, thereby allowing it to better differ-

entiate between the thematic subject of a lyric and the actual gender of its author.

To assess whether the performance gap between the two architectures was statistically significant, we performed a McNemar exact test on the paired test-set predictions. BERT significantly outperformed the TF-IDF linear baseline ($p < 0.001$). While the baseline correctly classified 368 songs missed by BERT, BERT successfully resolved 492 songs that baffled the TF-IDF model, proving that the deep learning approach brings a statistically significant improvement in capturing artistic gender indicators within lyrics.

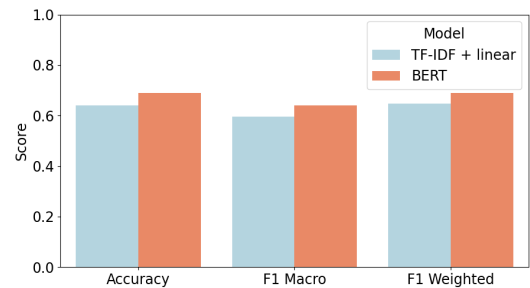


Figure 5: Performance metrics comparison (Accuracy, F1 Macro, and F1 Weighted) between the TF-IDF + linear baseline and the Fine-Tuned BERT model.

7 Feature Analysis

Contrary to the BERT model which operates as a black box that cannot be qualitatively interpreted, the TF-IDF pipeline allows us to better understand what the linear model explicitly learns. To do so, we inspect the most informative lexical features by gender (Figure 3).

This analysis shows that the model heavily relies on cross-gendered pronouns, such as *she* for the male artists and *he* or *him* for the female artists. Moreover, certain thematic domains appear to be heavily gendered. For instance, topics related to religion (*lord*) or generational transmission (*son*) seem to be predominantly associated with male singers.

Finally, one overarching narrative theme appears to be universally present in the lyrics of both classes: heterosexual romance. While male artists frequently refer to women (*she, girl, women, girl you*), female artists predominantly refer to men (*he, him, my man, boy you*). This pattern suggests that romance, structured through opposite-sex pronouns and direct address markers, stands out as one of the most discriminative and predictive themes for text-based gender classification.

8 Error Analysis

To look beyond aggregate statistical metrics, a comprehensive error analysis was conducted on the test set predictions. This qualitative review unveiled two systematic vulnerabilities in both architectures: the narrative perspective trap and the persistent impact of dataset imbalance.

First, both models rely heavily on surface-level pronouns and lexical markers to deduce author gender, making them highly susceptible to narrative shifts. For instance, in Murphy Elmore’s “*I kinda wanna thank him... push him up against the wall*” and Rockie Fresh’s “*...cause you know I don’t care about him*”, both models incorrectly predicted a female artist. This misclassification is likely due to the strong statistical association of the pronoun “*him*” with female-authored tracks. This failure mode proves that while both models effectively capture word-gender correlations, they lack the high-level semantic capability to track narrative perspective—specifically, distinguishing who is speaking from who is being spoken to.

Second, the per-artist error summary highlights how the male/female class imbalance penalizes minority class predictions. The top-ranking sources

of error are heavily dominated by prolific female vocalists, showcasing a mathematical majority bias where ambiguous tracks defaults to the male class.

9 Limitations

This study has several limitations that can be noticed. First, we didn’t use the genre of the songs where the lyrics were from, and the use of lyrics alone excludes vocal, musical, and performance-level cues that may be relevant to gender inference. Consequently, in particular, some high-weight terms of the TF-IDF model (Figure 3) appear to function as shortcuts that correlate with the target label rather than as robust semantic indicators, such as “*fuck*”, for example, which could be a word appearing more in rap music, and so mechanically appearing more in men’s lyrics, as there are more male rappers than female ones.

Second, the truncation of long lyrics to 512 tokens may remove informative sections, especially in genres with extended verses. Third, the binary gender formulation necessarily simplifies a more complex social reality and excludes non-binary identities as well as group performances, which was the majority of our initial dataset.

These limitations do not invalidate the study, but they do constrain the scope of the conclusions. The present work should therefore be understood as a controlled text classification experiment rather than a comprehensive account of gender in music.

10 Conclusion

We presented a controlled comparison between a sparse lexical baseline and a fine-tuned transformer model for predicting singer gender from song lyrics. Our results show that BERT outperforms the TF-IDF baseline on all main metrics and that this improvement is statistically significant.

Beyond performance, our analysis shows that both models are influenced by lexical shortcuts and narrative effects, while the transformer model is better able to exploit contextual information. Future work could extend this framework by incorporating acoustic features, metadata, or multi-modal representations to model gender and identity in music more comprehensively.

11 Contributions

Noam extracted the dataset we needed from the public Kaggle dataset using the API to extract the gender of the singer. He then fine-tuned Bert. He

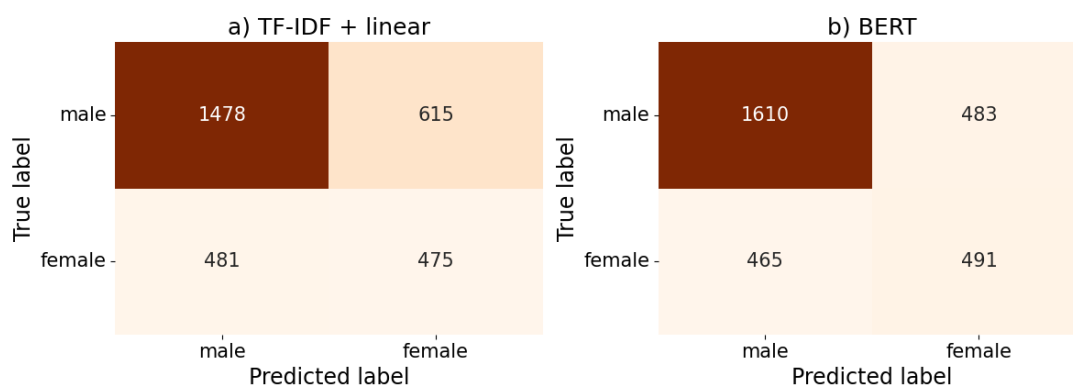


Figure 6: Confusion matrices on the test set: a) TF-IDF + linear baseline predictions vs. true labels, and b) BERT model predictions vs. true labels.

wrote the Results, the Feature Analysis and the Limitation parts.

Lelie filtered the original dataset by English language in the first notebook, coded the TF-IDF model in the second notebook and found a great part of the references. She wrote the Dataset and Task definition, Experimental Setup and Preprocessing parts.

Tabara coded the environment and import part in the second notebook, wrote the Abstract, Introduction, Error Analysis and Conclusion parts and implemented the article in Latex. We all coded various parts in the second notebook in collaboration, read literature on the subject and wrote the Model parts of the article.

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